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Effect of parametric uncertainty in numerical simulations of a hydrogen-fueled flameless combustion furnace using dimensionality reduction and non-linear regression

R. Amaduzzi [a,b,*], A. Procacci [a,b], A. Piscopo [a,b,c], R. Malpica Galassi [d], A. Parente [a,b,e,*]

[a] Aero-Thermo-Mechanics Laboratory, École Polytechnique de Bruxelles, Université Libre de Bruxelles, Brussels, Belgium
[b] Brussels Institute for Thermal-fluid systems and clean Energy (BRITE), Université Libre de Bruxelles and Vrije Universiteit Brussel, Brussels, Belgium
[c] Thermal Engineering & Combustion Unit, University of Mons, Mons, Belgium
[d] Mechanical and Aerospace Engineering Department, Sapienza University of Rome, Roma, Italy
[e] WEL Research Institute, Avenue Pasteur 6, 1300 Wavre, Belgium

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ABSTRACT

The objective of this work is to assess the propagation of the uncertainty in the 2D RANS model of a semi-industrial furnace, due to 9 uncertain input parameters. To collect the data required for this statistical analysis, Proper Orthogonal Decomposition combined with Gaussian Process Regression (POD/GPR) was employed to generate a surrogate model of a 2-dimensional Reynolds-Averaged Navier–Stokes (RANS) simulation in a 9-dimensional parameters space. The surrogate model is built by compressing the training data using POD, which reduces the dimensionality of the RANS response. GPR is then used for regression in the uncertain parameter space. We apply this methodology to a hydrogen-fueled semi-industrial furnace to assess the predictive uncertainty of the RANS simulations due to the uncertainty associated with turbulence, combustion and kinetics model input parameters. The results show that the POD/GPR surrogate model is able to accurately predict the temperature and species mass fractions in the furnace. We employ the surrogate to perform a global sensitivity analysis to determine the relative importance of the uncertain input parameters. It is found that the turbulence parameter $C_{2\epsilon}$ and the combustion model constant C_{mix} hold the strongest influence on the variability of the model response in the flame region, while the parameter controlling uncertainty of the inlet air mass flow rate is the one contributing the most in the recirculating region of the furnace. This study demonstrates the effectiveness of the POD/GPR approach in quantifying uncertainty in combustion systems and provides valuable insights into the contribution of different parameters to the response variance.

1. Introduction

The escalating global energy needs, coupled with the drive to curb emissions, has prompted extensive research and innovation in combustion technologies. These advancements aim to ensure reduced emissions and enhanced fuel adaptability, especially amid the ongoing shift toward renewable energy sources. Notably, the use of hydrogen as a clean energy carrier has been extensively investigated in a wide range of combustion applications in recent years.

Despite substantial advancements in computational fluid dynamics (CFD), their application in real-world industrial settings remains largely confined to Reynolds-averaged Navier Stokes (RANS) simulations. The practical use of large eddy simulations (LES) and direct numerical simulations (DNS) remains impractical in many cases due to their high computational expenses. In addition, the rapid growth of Machine Learning (ML) applications to forecast complex high-dimensional dynamics requires the correct characterization of data uncertainty. In their basic forms, ML methods indeed fail to provide any indication on the error of their predictions and its propagation.[1] Given this scenario, an effective approach to address the growing industrial need for cost-effective and dependable predictions involves enhancing the reliability of RANS by accurately assessing the uncertainties linked with numerical models. In particular, the estimation of the influence of non-deterministic input parameters on the quantities of interest (QoI)

* Correspondence to: Aero-Thermo-Mechanics Department. Université Libre de Bruxelles, Avenue F. D. Roosevelt 50, 1050 Bruxelles, Belgium.
E-mail addresses: ruggero.amaduzzi@ulb.be (R. Amaduzzi), alessandro.parente@ulb.be (A. Parente).

[1] Techniques such as Gaussian Process Regressions may provide an indication on the confidence of their regression's predictions. However, they are not a direct measure of the full-order model's error, but rather of the amount and distribution of the training data points and their correlation.

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is essential to correctly assess the reliability and robustness of the predictions.

In CFD applications, non-intrusive formulations of uncertainty quantification (UQ) techniques are generally preferred, as they allow to treat the CFD code as a black-box with given inputs and outputs. Generalized polynomial chaos (PC) has shown success in applications typically characterized by a low-dimensional input parameter space [1–3]. In combustion applications, PC techniques were already employed to propagate kinetics rate coefficients uncertainties [4], subfilter models' coefficients [5], scalar dissipation rate transport coefficients [6]. The combination of PC methods with compressed sensing and multifidelity techniques can alleviate the computational burden of sampling CFD simulations in high-dimensional uncertain parameters space: Huan et al. [7] performed a sensitivity analysis on a scramjet combustor including up to 24 uncertain parameters. However, such calculations still suffer from the known issue of the curse-of-dimensionality, requiring thousands of sampling points as the dimensionality increases. While techniques such as High Dimensional Model Representation (HDMR) [8] have been applied in literature to decrease the number of realizations required to build surrogate models, the sample size required by such models depends on the shape of the response surface and the magnitude of higher order interactions, which are unknown *a priori*. Without prior knowledge of the response of the system at hand, the large number of parameters generally determine the size of the sampling dataset. Proper orthogonal decomposition (POD), or principal component analysis (PCA), coupled with Gaussian process regression (GPR) have been used in literature to quantify uncertainty in scenarios involving many parameters. POD/PCA allows to significantly reduce the number of degrees of freedom of the model response, allowing to generate an efficient surrogate model of the full-order model and thus facilitating uncertainty propagation. This and other similar approaches have been successfully applied to a number of applications in engineering [9–12].

To the authors' knowledge, this methodology has never been employed for the UQ analysis of combustion systems in industrial-relevant conditions. In this work, we investigate the predictive uncertainty of RANS simulations due to the uncertainty associated with combustion, turbulence, kinetics model input parameters along with other operative parameters. The case study is the hydrogen-fired ULB flameless furnace [13,14]. The paper is organized in two parts. First, the POD/GPR model is validated against the training dataset with a *k-fold* cross validation. Second, the predictive uncertainty of RANS model is assessed and a global sensitivity analysis to the uncertain input parameters is provided.

2. Uncertainty propagation

2.1. POD/GPR surrogate model

High-dimensional uncertainty propagation and the following global sensitivity analysis require an efficient and accurate surrogate model of the expensive CFD simulations. In this work, the surrogate model is built by coupling POD for the dimensionality reduction, i.e., data compression, and GPR for the regression in the uncertain parameters space.

POD is a decomposition technique widely employed in fluid-dynamics to identify the most energetic structures in the flow, enabling dimensionality reduction [15]. Suppose to have p simulations obtained as realizations of the CFD model for p samples of the uncertain parameters $\boldsymbol{\theta}$; in the present case, each realization is a steady-state CFD simulation ran with the set of parameters $\boldsymbol{\theta}$. A data matrix $\mathbf{X} \in \mathbb{R}^{n,p}$ is assembled by stacking the p simulations $\mathbf{x}(\mathbf{r}, \boldsymbol{\theta})$ in the training dataset as column vectors of dimension $n = n_c \times n_f$, where n_c is the number of computational cells and n_f is the number of features (e.g., temperature and chemical species). Here, $\mathbf{r}$ denotes the spatial degrees of freedom. In the case of CFD simulations, in general we have that $n \gg p$. The data matrix can be also seen as the concatenation of n_f matrices $\mathbf{X}_i \in \mathbb{R}^{n_c,p}$, each one containing the data describing one feature.

Before applying the POD, it is necessary to scale the data matrix to avoid that features with low absolute variance, such as radical species, are relegated to low energy modes and thus discarded during the truncation. The scaling is achieved by removing the mean $\mu(\mathbf{X}_i)$ from each feature matrix $\mathbf{X}_i$, which is then divided by the standard deviation $\sigma(\mathbf{X}_i)$ such that $\mathbf{X}_{0,i} = [\mathbf{X}_i - \mu(\mathbf{X}_i)]/\sigma(\mathbf{X}_i)$. This ensures that each scaled feature matrix $\mathbf{X}_{0,i}$ has zero mean and unitary variance.

The POD is performed by applying a singular value decomposition (SVD) to the matrix $\mathbf{X}_0$:

$$\mathbf{X}_0(\mathbf{r}, \boldsymbol{\theta}) = \mathbf{U}(\mathbf{r})\boldsymbol{\Sigma}\mathbf{V}(\boldsymbol{\theta})^T, \tag{1}$$

where $\mathbf{U}(\mathbf{r}) \in \mathbb{R}^{n,p}$ and $\mathbf{V}(\boldsymbol{\theta}) \in \mathbb{R}^{p,p}$ are the orthonormal basis for the column space and row space of $\mathbf{X}_0$, respectively. The diagonal matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{p,p}$ contains the singular values of $\mathbf{X}_0$. In Eq. (1) we observe that POD effectively separate the spatial ($\mathbf{r}$) and parametric ($\boldsymbol{\theta}$) information.

The POD is generally selected for dimensionality reduction because the q-order truncation is guaranteed to minimize the l_2 norm reconstruction error. In other words, the matrix $\mathbf{X}_0$ can be approximated as $\mathbf{X}_0 \approx \mathbf{U}_q\mathbf{A}_q^T$, where $\mathbf{U}_q$ and $\mathbf{A}_q$ contain the first q columns of $\mathbf{U}$ and $\mathbf{A} = (\boldsymbol{\Sigma}\mathbf{V}^T)^T$, with $q < p$.

This means that instead of directly predicting the n-dimensional state $\mathbf{x}_0(\mathbf{r}, \boldsymbol{\theta}^*)$ in an unexplored design condition $\boldsymbol{\theta}^*$, we predict the compressed q-dimensional array $\mathbf{a}_q(\boldsymbol{\theta}^*)$, and project it back into the high-dimensional space:

$$\mathbf{x}_0(\mathbf{r}, \boldsymbol{\theta}^*) = \mathbf{U}_q(\mathbf{r})\ \mathbf{a}_q(\boldsymbol{\theta}^*). \tag{2}$$

Thus, the data compression operation allows us to deal with a much simpler surrogate construction problem of the type: $\mathbf{a}_q(\boldsymbol{\theta}) = f(\boldsymbol{\theta}) + \epsilon$, where the objective is to identify the function $f(\boldsymbol{\theta}) : \mathbb{R}^d \rightarrow \mathbb{R}^q$ that relates the values of the POD coefficients $\mathbf{a}_q$ with the uncertain parameters, in the presence of a Gaussian discrepancy $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma_n\mathbf{I})$ between the data and the surrogate model. This regression task is handled using the GPR model [16]. The GPR is generally employed to approximate scalar valued functions, i.e. functions of the type $f(\boldsymbol{\theta}) : \mathbb{R}^d \rightarrow \mathbb{R}$. Thus, to predict $\mathbf{a}_q$ we build q regression models, each one predicting a component of the low-dimensional array:

$$a_i = f_i(\boldsymbol{\theta}) + \epsilon_i \quad \text{for } i = 1, \dots, q. \tag{3}$$

The GPR is a Bayesian regression framework in which the regressing function f is assumed to be a sample from a Gaussian Process $f_i(\boldsymbol{\theta}) \sim \mathcal{GP}(m(\boldsymbol{\theta}), k(\boldsymbol{\theta}, \boldsymbol{\theta}'))$ [16], where $m(\boldsymbol{\theta})$ and $k(\boldsymbol{\theta}, \boldsymbol{\theta}')$ are the mean and covariance functions. In practice, the function f_i is modeled as a sample from a multivariate Gaussian distribution $\mathbf{f}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{K}(\boldsymbol{\Theta}, \boldsymbol{\Theta}))$, which represents the prior. The covariance matrix is computed by applying a kernel function k on the inputs such that $\mathbf{K}_{i,j} = k(\theta_i, \theta_j, \phi)$, where ϕ represents the length scale of the kernel. The posterior distribution is computed by applying Bayes' theorem:

$$\mathrm{p}(\mathbf{f}|\mathbf{a}_i, \boldsymbol{\Theta}) = \frac{\mathrm{p}(\mathbf{a}_i|\mathbf{f}, \boldsymbol{\Theta})\mathrm{p}(\mathbf{f}|\boldsymbol{\Theta})}{\mathrm{p}(\mathbf{a}_i|\boldsymbol{\Theta})}, \tag{4}$$

where $\mathbf{a}_i$ represents the observed values, or in our case the ith column of the matrix $\mathbf{A}_q$.

The hyperparameters of the GPR, i.e. the length scale of the kernel and the discrepancy parameter σ_n, are trained by minimizing the negative marginal log likelihood using a gradient-based method.

Finally, we obtain the predictive posterior distribution by computing the joint distribution between the training points $\boldsymbol{\Theta}$ and the prediction points $\boldsymbol{\Theta}_*$, and conditioning this distribution on the observed values.

Such a surrogate model of the stochastic response of the full order model allows us to efficiently propagate uncertainty and evaluate global sensitivity indices.

2.2. Global sensitivity analysis

Variance based methods for sensitivity analysis have proven to be versatile and effective. Given a model $Y = f(\theta_1, \theta_2, \dots, \theta_k)$, with Y a scalar (the QoI), the variance based first order effect for a generic parameter θ_k can be written as:

$$V_{\theta_k}(E_{\theta_{\sim k}}(Y|\theta_k)) \tag{5}$$

where θ_k is the kth parameter and $\theta_{\sim k}$ denotes the matrix of all factors but θ_k. The inner expectation operator $E_{\theta_{\sim k}}$ represents the mean of Y taken over all possible values of $\theta_{\sim k}$ while keeping θ_k fixed. The associated sensitivity measure, the first order sensitivity coefficient, or first order Sobol index [17,18], is:

$$S_k = \frac{V_{\theta_k}(E_{\theta_{\sim k}}(Y|P_k))}{V(Y)} \tag{6}$$

A second popular variance based measure is the total effect index [18]:

$$\begin{aligned} S_{T_k} &= \frac{E_{\theta_{\sim k}}(V_{\theta_k}(E_{\theta_{\sim k}}(Y|\theta_k)))}{V(Y)} \\ &= 1 - \frac{V_{\theta_{\sim k}}(E_{\theta_{\sim k}}(Y|\theta_{\sim k}))}{V(Y)} \end{aligned} \tag{7}$$

S_{T_k} measures the total effect, meaning the first and higher order interactions of θ_k are all taken into account. As explained in [18], S_k, S_{T_k} can be also interpreted in terms of expected reduction of variance: $V_{\theta_k}(E_{\theta_{\sim k}}(Y|\theta_k))$ is the expected reduction in variance obtained if θ_k could be fixed, and $E_{\theta_{\sim k}}(V_{\theta_k}(E_{\theta_{\sim k}}(Y|\theta_k)))$ the expected variance left if all factors but θ_k could be fixed. Saltelli's formulation [18] for the total effect indices S_{T_k} computation is employed in this work.

3. Application

3.1. RANS of the ULB flameless furnace

The methodology is tested on the ULB flameless furnace [13,14]. The furnace consists of a cubic combustion chamber (700 × 700 × 700 mm inner dimensions) and it is fired by a bottom-mounted WS REKUMAT M150 commercial burner equipped with an integrated heat exchanger to preheat the combustion air. The fuel is injected via a centrally located nozzle with internal diameter (ID) 8 mm and surrounded by a coaxial air jet with variable diameter to adjust the air entrainment. In-flame temperature measurements are acquired using an air-cooled suction pyrometer equipped with a 0.5 mm diameter B-type thermocouple. Fig. 1 shows the cross-section of the furnace and a sketch of the injector. Test case 3 (T3) from [14] is selected. T3 features 100% H_2, a nominal input power of 15 kW, a cooling power of 5.1 kW, an equivalence ratio $\varphi = 0.8$ and an air ID of 16 mm. Temperature was measured over 6 radial positions at 8 different distances from the injector, for a total of 48 probe locations.

To reduce the computational cost, 2D RANS simulations of the furnace, with 43700 cells, were performed using the commercial software Ansys Fluent 19.3 [19]. Previous studies [13] proved the relative accuracy of 2D RANS simulations compared to their 3D counterparts on the same system.[2] Turbulence was modeled by means of the realizable $k-\varepsilon$ model, while the PaSR combustion model was employed to model turbulence-chemistry interactions [20,21]. A global mixing time scale, fraction (governed by the model constant C_{mix}) of the integral time scale $\tau_I = k/\varepsilon$ was employed, while the characteristic chemical time scale was determined from the formation rates. The global mixing timescale reads:

$$\tau_{mix} = C_{mix}\frac{k}{\varepsilon}, \tag{8}$$

[2] The temperature error is shown to be below 10% for a given C_{mix} value between 3D and 2D simulations of the same case, which is an acceptable value for the purposes of this work.

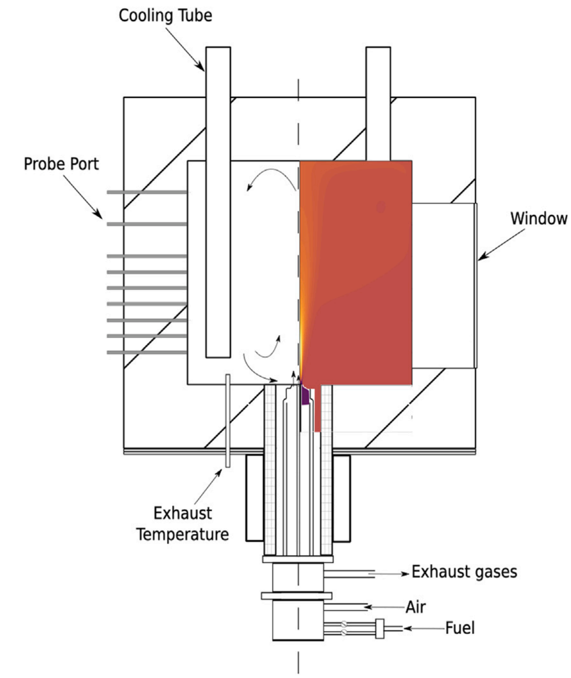

Fig. 1. Cross-section of the ULB furnace, with a qualitative sketch of the mean temperature field. For the exact probe locations, see Fig. 5.
Source: Adapted from [14].

where k is the turbulent kinetic energy and ε its dissipation rate. The H_2 core mechanism and the corresponding NOx sub-mechanism from the C1–C3 mechanism from Polimi [22,23] were used. Radiation was modeled with the Discrete Ordinates model, employing the Weighted-Sum-of-Gray-Gases absorption/emission model. A description of the uncertain input parameters for each model is presented in the next section.

3.2. Uncertain parameters definition

To take into account as much modeling and parametric uncertainty as possible, 9 uncertain parameters have been chosen, including reaction rates constants, turbulence and combustion models parameters and inlet mass flow variations. With regards to kinetic rates uncertain parameters selection, a computational singular perturbation (CSP) analysis [24] of 1D flamelet solutions was carried out to determine the reactions mostly contributing to the temperature dynamics. Reactions R3, R5 and R16 of the mechanism were selected (Table 1). The characterization of their uncertainty was carried out using the Uncertainty Factor (UF) defined as:

$$\mathrm{UF} = \log_{10}\left(\frac{A_{\max}}{A_0}\right). \tag{9}$$

The UFs of the select reactions were taken from [25]. The reactions and their respective UFs are listed in Table 1. We assume that the pre-exponential factor is log-normally distributed. Therefore, we employ normal distributions for $\ln(A)$. We compute the distribution standard deviation as:

$$\sigma = \frac{\ln\left(10^{UF}\right)}{3}, \tag{10}$$

having considered $A_{\max}$ as the 3σ value of the normal distribution of $\ln(A)$. More informations on the CSP analysis and the choice of the important reactions can be found in the supplementary material.

Three uncertain model constants were taken into account for the realizable $k-\varepsilon$ [26]: $C_{2\varepsilon}$, σ_k and σ_ε. Given the lack of knowledge about their prior distribution, a uniform distribution was assigned to

Table 1
Important reactions and their respective uncertainty factor UF.

Reaction		UF	Reference
(R3)	$H_2 + OH \leftrightarrow H + H_2O$	0.3	Konnov 2008 [25]
(R5)	$H + O_2 \leftrightarrow O + OH$	0.18	Konnov 2008 [25]
(R16)	$H_2 + O_2 \leftrightarrow H + HO_2$	0.3	Konnov 2008 [25]

Table 2
Uncertain parameters and their uniform prior distributions.

Parameter	Lower bound	Upper bound
$C_{2\epsilon}$	1.5	3
σ_k	0.8	1.4
σ_ϵ	0.8	1.4
C_{mix}	0.01	0.6
$\dot{m}_f$ [kg/s]	$1.211 \cdot 10^{-4}$	$5.083 \cdot 10^{-3}$
$\dot{m}_{air}$ [kg/s]	$1.290 \cdot 10^{-4}$	$5.569 \cdot 10^{-3}$

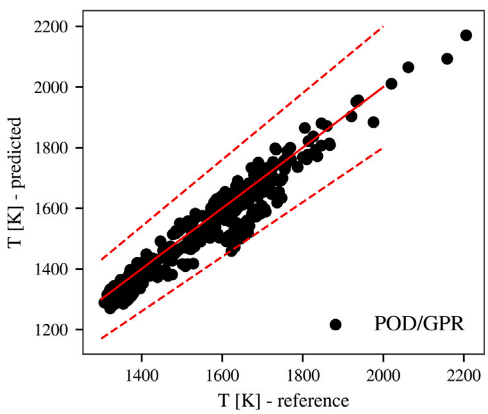

Fig. 2. Predicted temperature parity plot for the best performing *k-fold* cross validation at probes locations. Dashed lines represent an error of 10%.

the turbulence model parameters, selecting the lower and upper bounds from the literature review in [27]. Typical values for the C_{mix} constant in the PaSR combustion model in literature range from 0.01 to 0.6. As for the turbulence model, a uniform distribution was assigned to the C_{mix} parameter. Lastly, inlet fuel and air mass flows uncertainties were taken into account following the experimental uncertainty reported by [14]. Uniform distributions were assigned to both parameters. Table 2 summarizes the chosen uncertain parameters and their distributions.

The multivariate joint distribution of the 9 parameters was then sampled using Latin Hypercube Sampling (LHS) for a total of 128 evaluation points. For each one of those, a CFD simulation was carried out. To determine the sampling points from the multivariate distribution, the Python library ChaosPy [28] was employed. Then, the POD/GPR surrogate model was built using the in-house Python library OpenMEASURE [29].

4. Results

Once fitted, the POD/GPR surrogate model allows to compute the approximate QoI responses as functions of the uncertain parameters. The output QoIs considered in this study are temperature T, and species mass fractions of H_2O and OH. To mimic the averaging effect that the application of suction pyrometer (see Section 3.1) implies for experimental temperature measurements, the CFD results and the POD/GPR reconstructed fields were averaged in a volume defined by a sphere with radius 10 mm (the suction pyrometer diameter) to allow a fair comparison.

The accuracy of the surrogate model is assessed via *k-fold* cross validation. The 128 simulations dataset is shuffled and split into 14 groups (folds); for each group 8 simulations are taken as test points, while the rest are used for training. Fig. 2 shows the parity plot for the fold that yields the best R^2 score for temperature predictions on the test points, $R^2 = 0.9287$. Prediction errors are well within the 10% limits. Table 3 lists the minimum, maximum values for the mean RMSE (and its standard deviation) and R^2 for the *k-fold* cross validation for the three QoIs in the kth fold, along with the average RMSE and R^2 across all folds.

Table 3
Minimum, maximum and average-across-folds values for the QoI. RMSE_{mean} represents the mean RMSE in the kth fold across the test points while RMSE_{std} is the respective standard deviation.

		$RMSE_{mean}$	$RMSE_{std}$	R^2
T [K]	min	40.49	20.07	0.7632
	max	85.39	71.64	0.9288
	avg	64.11	13.54	0.8572
H_2O	min	0.002730	0.001256	0.8264
	max	0.013021	0.016556	0.9801
	avg	0.006639	0.003989	0.8572
OH	min	$1.9924 \cdot 10^{-4}$	$1.0027 \cdot 10^{-4}$	0.8160
	max	$4.5640 \cdot 10^{-4}$	$3.5694 \cdot 10^{-4}$	0.9556
	avg	$3.5577 \cdot 10^{-4}$	$7.4832 \cdot 10^{-5}$	0.9026

Table 4
Average-across-folds values of the RMSE of the reconstruction error ($\mathrm{RMSE}_{\mathrm{rec}}$) and the prediction error ($\mathrm{RMSE}_{\mathrm{pred}}$).

	$RMSE_{\mathrm{rec}}$	$RMSE_{\mathrm{pred}}$
T [K]	8.670	64.11
H_2O	0.0007986	0.006639
OH	$8.852 \cdot 10^{-5}$	$3.5577 \cdot 10^{-4}$

Given the high dimensionality of the parameter space and the relatively low number of simulations, the prediction errors are remarkably low. Such figures of merit could not be obtained using Polynomial Chaos Expansions (PCE) [30] fitted on the data with L1-regularized regression methods. Quadrature methods for estimating the PCE coefficients were not even applicable given the too high dimensionality of the parameter space, since 217, 1843 and 12 805 simulations are needed in a 9-dimensional space for 2nd, 3rd and 4th order sparse quadrature, respectively.

To assess the impact of the POD reconstruction error on the overall prediction error, the reconstruction error has been estimated using the same k-fold cross-validation method explained above. The test simulations are reconstructed using the following methodology:

$$\mathbf{X}^*_{0,\mathrm{rec}} = \mathbf{U}\mathbf{U}^T\mathbf{X}^*_0, \tag{11}$$

where $\mathbf{X}^*_0$ is the matrix containing the test simulations centered and scaled using the mean and standard deviation of the training dataset. Due to the truncation, the matrix $\mathbf{U}\mathbf{U}^T$ is slightly different from the identity matrix resulting in an error between the observed and reconstructed simulations.

Table 4 shows the comparison between the average-across-folds reconstruction and prediction errors. For all the features, the reconstruction error is one order of magnitude smaller than the global prediction error, meaning that the truncation error has little to no impact on the prediction error of the surrogate model.

Fig. 3 shows the effect of the training dataset size, i.e., the number of simulations used to train the POD/GPR model, on the regression accuracy. The error bands show the standard deviation of the root mean square error of the mean GPR predictions calculated across the k-folds. The almost steady trend for H_2O error is due to the low variability of the water vapor field in the considered dataset. Since H_2O is a final product of combustion, it is not greatly affected by the considered parameters if the oxidation is complete. For this reason, the POD/GPR model prediction RMSE does not decrease if the number of training simulations increases. For temperature and OH mass concentration, we observe an acceptable convergence of statistics with 120 simulations.

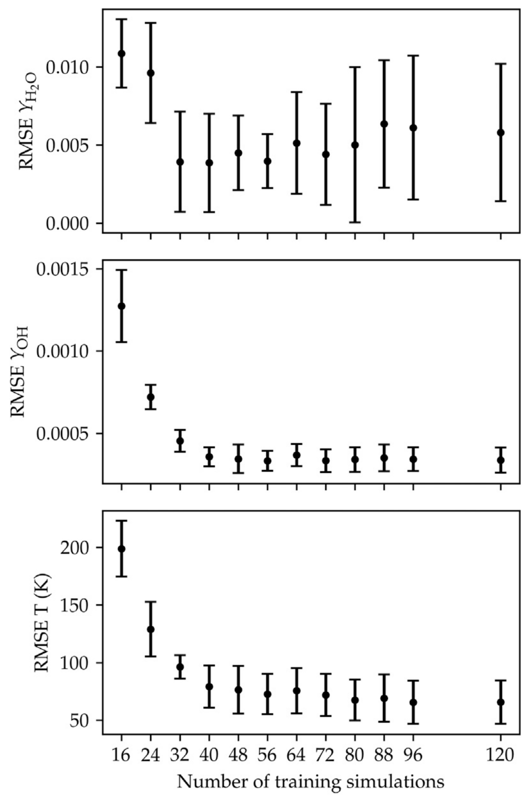

Fig. 3. Effect of the training dataset size on the propagated uncertainty.

4.1. Uncertainty propagation

Once the surrogate model is available, samples of the QoI response can be obtained at negligible cost. Therefore, Sobol samples of the input parameter space distribution were drawn and QoI responses were computed with the POD/GPR model to determine the temperature probability density function (PDF) in each experimental probe location. The experimental dataset consists of 8 axial locations, each with 6 radial points, for a total of 48 probing locations. The uncertainty in the temperature probe locations is shown in Fig. 4.

We observe that the predicted temperature uncertainty bands are visibly greater along the axis of the combustion chamber, where temperature is higher. On the contrary, as we move in the radial direction, the overall variance of the predictions decreases. While the "culprits" of the predictive uncertainty will be revealed in the next section, it is interesting to notice that even taking into account the 25th–75th percentiles, on the axial locations the propagated uncertainty spans ~400 K in the worst case, indicating that the model is highly sensible to the choice of the considered model parameters. While this behavior may be intuitively expected, here we quantify such sensitivity. Depending on the choice of input parameters, the model may yield a huge range of responses, spanning from cold flows to great overestimation of temperature.

4.2. Sensitivity analysis

To rank the effect of the uncertain parameters on the model response, we analyzed the response variance using the Sobol indices. The variance-based indices convergence was assured via a progressively increasing Sobol sampling of the parameters joint distribution. The surrogate model was then evaluated in 16 384 sampling points to determine the first and total order Sobol indices following [18]. Fig. 5 shows the total order Sobol indices in the probing locations in the

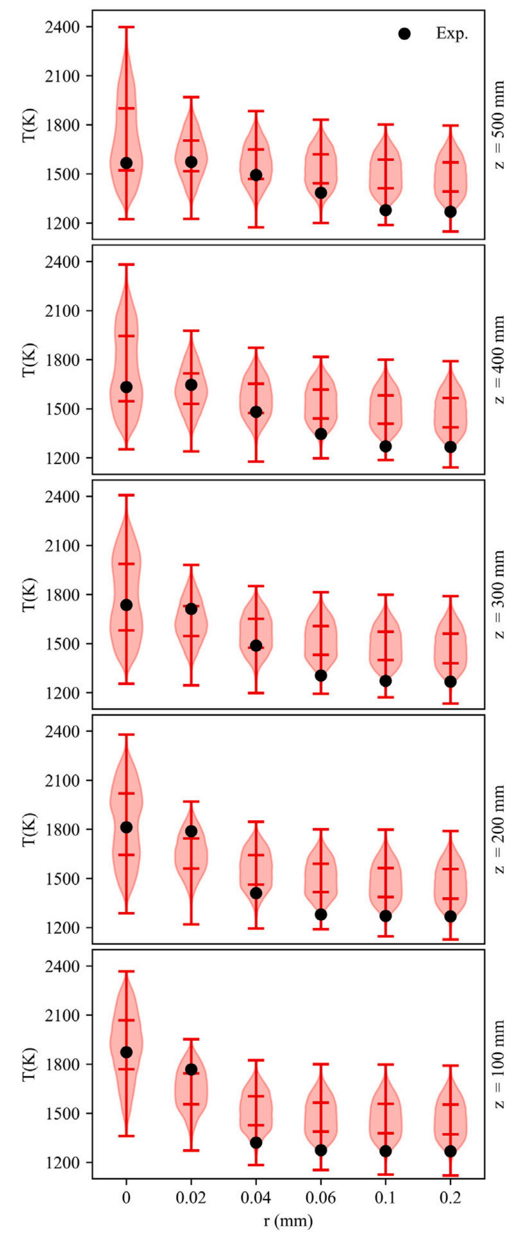

Fig. 4. Uncertainty propagation for temperature predictions along 5 different axial locations. Ticks inside of the PDF represent the 25th and 75th percentiles, respectively. Black dots represent experimental measurements from [14] (reported measurement error is 5% of measured value).

combustion chamber. Each circle marker represents the variance share in that axial/radial position. The radius of the marker represents the sum of the three Sobol indices S_{T_k}. The Sobol indices analysis of the other two QoIs, H_2O and OH, are supplied within the supplementary material.

Overall, three parameters account for 98.7% of the total variance: the PaSR C_{mix} constant, the realizable $k-\varepsilon$ $C_{2\varepsilon}$ constant and the air mass flow $\dot{m}_{air}$. In particular, the first two parameters alone account for 91.4%. Along the furnace axis, the two most important parameters are the $C_{2\varepsilon}$ and C_{mix} constants. The relative impact of $C_{2\varepsilon}$ increases with the distance from the injector, as combustion mostly takes place in the injector-adjacent region of the combustion chamber. $C_{2\varepsilon}$ is the parameter in the turbulence model that is related to the rate of destruction of the turbulent kinetic energy ε, and it is correlated to the penetration of the fuel jet. Its relative importance therefore

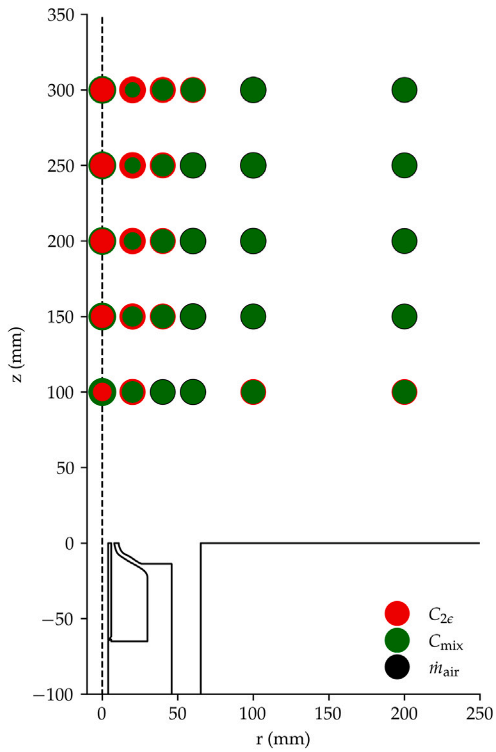

Fig. 5. Total order Sobol indices of the most impactful parameters at probe locations. In each location, the radius of the circle marker represents the sum of the two highest impacting Sobol indices S_{T_k}.

increases as the jet propagates into the chamber. Far from the axis, in the recirculating region of the furnace, the contribution of the turbulence model parameters fades and the highest contribution to the total response variance is given by the C_{mix} and $\dot{m}_{air}$ parameters. As mentioned, the C_{mix} constant determines the mixing timescale for the combustion model, which influences the ignition location of the flame in the case of mixing-controlled oxidation. m_{air} is the air mass flow rate, which controls the recirculation ratio in the furnace. This region is characterized by strong recirculation of hot combustion products which then get entrained into the main inlet jet. As explained in [13,14], variations in the air inflow strongly affect the recirculation pattern and therefore the structure of the flame inside the furnace, and the distribution of the Sobol indices corroborates the literature trends. A detailed table reporting the three main parameters' first and total order Sobol indices in the temperature probe locations, along with their difference, can be found in the supplementary material.

The kinetic rates uncertainties globally account for only 0.8% of the total variance. While the uncertainty of the H_2 core mechanisms is generally well known in the case of canonical reactors (as reflected by the UF of most reactions [31]), its propagation in a system of this scale has not been investigated in literature. Similarly, in this case, literature trends are somewhat confirmed, as the use of different kinetic schemes were found to have little influence on the model predictions in previous works on the same system [14].

5. Conclusions

Proper orthogonal decomposition combined with Gaussian process regression (POD/GPR) was employed to compute the uncertainty propagation of 9 uncertain parameters that characterize 2D RANS modeling of a hydrogen fueled semi-industrial furnace. The POD/GPR surrogate model was fitted starting from 128 simulations stemming from Latin hypercube sampling of the joint distribution of the 9 input parameters. The uncertainty was then propagated to the predictions of temperature, showing how the model is highly sensitive to the chosen parameters. The highest uncertainty is found along the central axis of the furnace, where the propagated temperature uncertainty spans as much as ∼400 K. A variance-based global sensitivity analysis of the surrogate model response was then employed to assess the relative influence of each parameter on the global variance. It was found that the turbulence parameter $C_{2\epsilon}$ and the PaSR combustion model constant C_{mix} hold the strongest influence on the variability of the model response in the flame region of the furnace. Moving far from the central axis, the uncertainty on the inlet air mass flow $\dot{m}_{air}$ becomes more important, as the air inlet jet strongly affects the recirculation pattern of the combustion chamber. More in general, we proved that the POD/GPR approach's effectiveness stems from its ability to navigate a high-dimensional space using a relatively small number of simulations. This enables the preliminary assessment of numerous uncertain parameters in a relatively simple model like 2D RANS. Subsequently, the impact of only the most crucial parameters will be evaluated using more sophisticated models, such as 3D RANS or LES, in future work. With this in mind, the natural continuation would lie in the implementation of multi-fidelity approaches and backward uncertainty propagation to better characterize the numerous uncertain inputs of the models.

Novelty and significance statement

The transition towards carbon-free combustion requires the retrofitting of existing technologies to different fuels. In this context it is crucial to assess predictive uncertainty in order to have reliable models. The quantification of uncertainty due to model parameters plays a key role in numerical predictions of combustion systems, especially when low-fidelity models such as RANS are employed. UQ methods are notably expensive, especially when the dimensionality of the uncertain parameter space is large. In this work, we propose to use the POD/GPR technique to build a surrogate model of the 2-dimensional RANS response as a function of 9 uncertain parameters, among which we included reaction rates constants, turbulence and combustion models parameters and inlet mass flow variations. The surrogate model allowed us to draw important conclusions on the contribution of such parameters to the response variance. It was found that the turbulence parameter $C_{2\epsilon}$ and the PaSR combustion model constant C_{mix} hold the strongest influence on the variability of the model response in the flame region of the furnace. Moving far from the central axis, the uncertainty on the inlet air mass flow u_{air} becomes more important, as the air inlet jet strongly affects the recirculation pattern of the combustion chamber. More in general, we proved that strength of the POD/GPR approach lies in its capacity to maneuver through a complex space with relatively few simulations compared to other typical UQ approaches. This allows for an initial evaluation of multiple uncertain parameters in a simpler model like 2D RANS. Then, the more critical parameters can be scrutinized using advanced models like 3D RANS or LES.

CRediT authorship contribution statement

R. Amaduzzi: Designed research, Implemented the algorithms, Performed research, Analyzed data, Wrote the paper. **A. Procacci:** Designed research, Implemented the algorithms, Performed research, Analyzed data, Wrote the paper. **A. Piscopo:** Performed research, Analyzed data, Wrote the paper. **R. Malpica Galassi:** Designed research, Supervised research, Wrote the paper. **A. Parente:** Supervised research, Wrote the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Additional analysis and data mentioned in the paper is available in the supplementary material.

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